SDV Resource Governor (sdv-resource-governor)

Deterministic 8-class physical resource governor for a Software-Defined Vessel (SDV). Evaluates payload admission & placement over a 14-day mission envelope (Scenario A / D1), executes live operational arbitration under safety deadlines (Scenario B / D3), enforces graceful mode step-down and load shedding under generator or cooling degradation (Scenario B / D4), and models crew regime transitions across 14, 6, and 0 personnel (Scenario C).

1. Where the Engine Decides in Each Execution Mode

Per Gate B1, a UI badge alone never proves remote execution. The governor explicitly separates and measures where the solver executes:

* 1. Standalone Laptop Mode (MacBook Pro M1 Pro, arm64, macOS):
  * Runs 100% of unit/integration/adversarial tests, 1 Hz deterministic event-driven simulation, 90-second compressed Scenario B replay, and the stateful single-screen 2D operator console (http://127.0.0.1:8765) without requiring any external board connection.
  * Records host, architecture, PID, RSS memory, wall-clock execution time (time.perf_counter_ns), source tree SHA-256, and canonical semantic_sha256.
* 2. Remote Edge Linux SBC Mode (ssh ardu -> Arduino Ventuno Q / training06, Qualcomm QCS8300 Monza 8-core aarch64, 14 GiB RAM, Ubuntu 24.04.4 LTS):
  * Deploys the exact same pure-Python governor package (src/sdv_governor/) and declarative YAML manifests (data/) into /home/arduino/sdv-resource-governor/.venv.
  * Executes the solver natively on the aarch64 Linux processor (10.0.0.130), capturing remote hostname, PID, machine, rss_kib, wall_elapsed_ms, code_sha256, and verifying bit-for-bit identical semantic_sha256 against the MacBook Pro (DEP-02, DEP-03).
  * If the board is physically unplugged or unpowered on USB-C/LAN, sdv-governor doctor and deploy-board report BLOCKED_CONNECTION cleanly without halting local execution.
* 3. STM32H5 MCU Scope:
  * Explicitly excluded from the admission/arbitration decision loop; no motor-control sketch is flashed or required for the challenge.

2. Quickstart (Clean Laptop or Linux Board)

```
# 0. Install in editable or standard wheel mode (offline-safe)
pip install --no-build-isolation --no-deps -e .

# 1. Validate all 6 guest manifests (G1..G6), ship profile, mission envelope, scenarios, priority profiles, and errata
python -m sdv_governor.cli validate

# 2. Check local & remote hardware readiness (never fails if board is disconnected)
python -m sdv_governor.cli doctor

# 3. Evaluate all 5 combinations of 4 official guests (Scenario A mass & placement regression)
python -m sdv_governor.cli combinations

# 4. Run 3 consecutive Scenario B replays and verify identical semantic SHA-256
python -m sdv_governor.cli replay --runs 3

# 5. Run the complete acceptance & regression test suite (98 tests: unit + integration + adversarial) and emit JUnit XML
pytest -v --junitxml=artifacts/final_run/junit.xml

# 6. Generate and fail-closed verify an evidence bundle (plus all 28 adversarial mutation checks)
python -m sdv_governor.cli campaign --run-id release_s6 --junit-xml artifacts/final_run/junit.xml --with-remote-parity
python -m sdv_governor.cli verify-evidence artifacts/release_s6 --run-mutations

# 7. Launch the Stateful Single-Screen 2D Operator Console (http://127.0.0.1:8765)
python -m sdv_governor.cli serve-ui --port 8765
```

3. Eight Physical Resource Classes & Key Physical Invariants

Resource Class	Nominal / Degraded Envelope	Enforced Physical Invariants
1. Power	200 kW cruise reserve (100 kW gen loss); LINE_A_FWD <= 150 kW, LINE_B_AFT <= 150 kW	Net of hotel load (never double-counted). Distinguishes instantaneous actual load vs firm committed load (G3 25 kW recharge reservation + G4 35 kW D3 recovery reservation).
2. Cooling	160 kW @ 12 deg C linear to 120 kW @ 30 deg C; -40% loop loss -> 72 kW @ 30 deg C	1:1 electrical-to-thermal coupling. Thermal inertia (+30 kW / 60 s = 1.8 MJ, 600 s cooldown) is stepped at 1 Hz and never used for sustained loads (>60 s).
3. Positions	Bay1 (<= 8,000 kg), Bay2 (<= 4,000 kg), Hangar (<= 1,000 kg, airframe <= 150 kg), Hangar Roof, Flight Deck, Stern Ramp	Powered-off / shed guests retain their physical bay, mast slot, mass, and stability footprint (PHY-08). hangar_roof enforces G4 VTOL drone clearance.
4. Mast	1 guest-allocatable slot (mast_3; mast_1, mast_2 fixed ship); <=60 kg, <=0.60 m2, air <=2 kW	No chilled-water manifold on mast. G1 (45 kg, 1.5 kW air) and G5 antenna (25 kg, 1.2 kW air) compete for the single free guest mast slot (1/1).
5. Mass & Stability	10,000 kg mass margin; 12,000 kg-eq stability (x1.0 deck, x1.2 hangar/roof, x3.0 mast)	Computed per subcomponent & position (G5 = 3000x1.0 + 25x3.0 = 3,075 kg-eq on mast_3 or 3,030 kg-eq on hangar_roof). Errata: G4 = 140 + 850 = 990 kg; 3 of 5 4-guest sets exceed 10,000 kg.
6. Data Links	Dual 10 Gbps LAN (10 Gbps single-ring guaranteed), 2-8 Mbps SATCOM (700 ms), RF LOS, Acoustic (15 s hold)	200 ms urgent commands never routed solely over 700 ms SATCOM. EMCON silence suppresses RF/SATCOM transmissions (G4 standby 0 bps).
7. Compute Rack	4 nodes (32 cores / 128 GiB / 1 accel each), 20 TB storage; 80% allocatable	25 cores, 102,400 MiB RAM, 1 whole indivisible accelerator per node (max 3 cluster-wide), 16,000 GB storage. Separate from Mac/Ventuno host RAM.
8. Time	D1 <= 30 s, D3 <= 2 s, D4 first shed <= 5 s / full <= 60 s, STOP/HOLD/SECURE <= 200 ms	Measured via time.perf_counter_ns after full 8-class budget evaluation; separated from canonical semantic_sha256.

4. Adding a New Guest Card (G7 Executable Example)

Any new payload card must conform to schemas/guest_manifest.schema.json and pass parse_guest_manifest without modifying solver code. Save a YAML manifest (or submit via POST /api/admit_live) with schema_version: "sdv-guest-manifest/1.0", 1:1 thermal coupling (cooling_water_w + cooling_air_mast_w == sustained_power_w), monotonic power hierarchy (min <= nominal <= sustained <= peak), and explicit value_origin provenance:

```
import json, yaml, jsonschema
from pathlib import Path
from sdv_governor.manifest import parse_guest_manifest

g7_yaml = """
schema_version: "sdv-guest-manifest/1.0"
guest_id: "G7"
name: "G7 -- Compact Expendable Bathythermograph (XBT) Launcher"
category: "AUXILIARY_PAYLOAD"
safety_critical: false
default_mode: "STANDBY"
can_buffer_offline: true
subcomponents:
```

```
  - subcomponent_id: "G7_DECK_LAUNCHER"
    part_of: "G7"
    name: "Stern Deck XBT Autolauncher Pod"
    mass_kg: 180
    footprint_m2: 1.2
    wind_area_m2: 0.8
    preferred_positions: ["bay1_b", "bay2"]
    allowed_positions: ["bay1_b", "bay2", "bay1_a"]
    cooling_route: "CHILLED_WATER_LOOP"
    requires_water_manifold: false
    value_origin: "ASSUMED"
modes:
  STANDBY:
    description: "Launcher indexed and heater idle"
    min_power_w: 200
    nominal_power_w: 300
    sustained_power_w: 300
    sustained_duration_s: 3600
    peak_power_w: 500
    peak_duration_s: 10
    cooling_water_w: 300
    cooling_air_mast_w: 0
    compute_cores: 1
    compute_ram_mib: 1024
    compute_accelerators: 0
    shared_storage_gb: 20
    backbone_lan_bps: 1000000
    satcom_bps: 0
    safety_critical: false
    uninterruptible: false
    min_done_duration_s: 0
    min_cooldown_s: 0
    forbids_emcon_silence: false
    requires_deck_clear: false
    requires_quiet_acoustic_state: false
    capability_retention_ratio: 0.25
    value_origin: "ASSUMED"
installation:
  port_only: true
  min_crew_required: 2
  install_time_min: 30
  remove_time_min: 20
  sea_install_max_wind_kn: 0
  sea_install_min_speed_kn: 0.0
  sea_install_max_speed_kn: 0.0
vehicle_replacement:
  supported_at_sea: false
  min_crew_required: 0
  swap_duration_min: 0
  min_speed_kn: 0.0
  max_speed_kn: 0.0
  max_wind_kn: 0
  max_sea_state: 0
ship_state_constraints:
  min_speed_kn_for_operation: 0.0
  max_speed_kn_for_operation: 20.0
  max_wind_kn_for_operation: 35
  max_sea_state_for_operation: 6
  requires_heading_into_wind_for_launch_recovery: false
  forbids_deck_machinery_while_active: false
  acoustic_hold_handshake_min_s: 0
  acoustic_hold_handshake_max_s: 0
  acoustic_hold_default_s: 0
  low_battery_threshold_pct: null
  emergency_landing_deadline_s: null
  max_flight_endurance_min: null
dependencies:
  requires_guests: []
  excludes_guests: []
  conditional_excludes: []
mission_capabilities:
  hydro_sampling_value: 20.0
"""
raw = yaml.safe_load(g7_yaml)
schema = json.loads(Path("schemas/guest_manifest.schema.json").read_text())
jsonschema.validate(raw, schema)
manifest = parse_guest_manifest(raw)
```

5. Repository Structure & Governance Traceability

* src/sdv_governor/: Pure Python deterministic engine (models.py, manifest.py, ship_state.py, budgets.py, placement.py, explanations.py, admission.py, arbitration.py, degradation.py, scheduling.py, replay.py, evidence.py, deployment.py, server.py, cli.py, pdf_export.py).

* schemas/ & data/: Strict JSON Schemas and declarative YAML manifests (vessel_baseline.yaml, mission_envelope_a10.yaml, G1..G6, survey_primary.yaml, rescue_primary.yaml, scenarios, and brief_errata.yaml).

* BACKGROUND_IP.md, AI_USE_DISCLOSURE.md, THIRD_PARTY_COMPONENTS.csv, ASSUMPTIONS.yaml, requirements_matrix.csv, hardware_inventory.json: Full provenance and compliance records.